NATIONAL ENGINEERING SCHOOL OF TUNIS

Department of Information and Communication Technologies

Report of the Internship Work

DEVELOPMENT OF A SMART TOOL TO MONITOR THE CSR ENVIRONMENTAL ACTIVITIES

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Abstract

This project presents the design and implementation of a **smart environmental monitoring system** developed as part of Corporate Social Responsibility (CSR) initiatives. The proposed solution integrates **IoT-based sensing technology** to measure key environmental indicators such as water flow, waste level, temperature, humidity, and energy consumption, providing real-time monitoring and data visualization through **Grafana dashboards**. An **AI-powered Optical Character Recognition (OCR)** module was developed to automatically extract relevant data from electricity and water invoices, significantly reducing manual processing time and human error. A **Role-Based Access Control (RBAC)** system ensures secure and personalized access for different user roles (Employee, Manager, and Top Manager). Field visits to enterprises, as well as to **SONEDE** and **STEG**, helped establish national benchmarks and consumption thresholds for performance comparison. The results confirm the stability, accuracy, and scalability of the proposed system, demonstrating its potential to support organizations in achieving **Sustainable Development Goals (SDGs)** through data-driven environmental monitoring and reporting.

Résumé

Ce projet présente la conception et la mise en œuvre d'un **système intelligent de suivi environnemental**, développé dans le cadre de la **Responsabilité Sociétale des Entreprises (RSE)**. La solution proposée repose sur des **capteurs IoT** permettant de mesurer et de surveiller en temps réel des indicateurs environnementaux clés tels que le débit d'eau, le niveau de déchets, la température, l'humidité et la consommation énergétique, avec visualisation via des **tableaux de bord Grafana**. Un module d'**extraction automatique des données à partir de factures** d'eau et d'électricité a été développé en utilisant une approche d'**Intelligence Artificielle basée sur la reconnaissance optique de caractères (OCR)**, afin d'améliorer la précision et d'automatiser la collecte des données. Un système de **contrôle d'accès basé sur les rôles (RBAC)** a également été intégré pour garantir la sécurité et la personnalisation de l'accès selon les profils (Employé, Manager, Top Manager). Des **visites d'entreprises**, ainsi que des rencontres avec la **SONEDE** et la **STEG**, ont permis de recueillir des données de référence sur les seuils de consommation et les stratégies nationales liées aux **Objectifs de Développement Durable (ODD)**. Les résultats obtenus démontrent la fiabilité et l'efficacité du système, confirmant son potentiel pour accompagner les organisations dans leur transition vers une gestion environnementale plus durable et intelligente.

List of Acronyms

CSR *Corporate Social Responsibility*
KPIs *Key Performance Indicators*
KGIs *Key Goal Indicators*
SDGs *Sustainable Development Goals*

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General Introduction

In a global context where Corporate Social Responsibility (CSR) has become a major strategic priority, the integration of innovative tools to monitor and optimize environmental and social activities is now essential. This internship project, conducted at SMART KOKUSAI, focused on the design and development of a smart tool for monitoring CSR activities.

In Tunisia, the rise of digital technologies offers new opportunities to strengthen CSR practices and promote transparency. In this perspective, the proposed solution was designed to be both user-friendly and innovative, encouraging active participation from employees and communities while enabling companies to measure the impact of their social initiatives in alignment with international standards.

This internship was carried out within the framework of a broader project aimed at improving industrial processes while ensuring compliance with environmental and social standards. Its main objective was to develop innovative solutions that help monitor and optimize company activities with a positive impact on society and the environment.

My role consisted in contributing to the design and development of a smart monitoring system based on Internet of Things (IoT) technologies. The system was intended to collect and analyze key performance indicators in real time. This experience enabled me to face real professional challenges, collaborate effectively within a team, and contribute to a project with tangible impact.

This report presents the work accomplished during this internship and is structured into four chapters :

1. The first chapter provides a general overview of the project, the host company's context, and the internship objectives.
2. The second chapter presents the state of the art related to CSR, the

Sustainable Development Goals (SDGs), and the IoT technologies supporting the developed solution.

3. The third chapter details the design of the smart monitoring system, including its technical and functional architecture.
4. Finally, the fourth chapter describes the implementation, testing, and evaluation of the developed prototype.

Chapitre 1

Project Overview

1.1 Introduction

This chapter presents the host company by outlining its profile and organizational struc- ture. It also identifies the requirements that motivated the development of the tool, ensuring its alignment with the company's operations and objectives.

1.2 Profile of the Host Company

SMART KOKUSAI Ltd is a recently established small company, in April 2022, in order to provide a modern and innovative engineering services and RDI researches. The company is locatedinAriana,Tunisia. Thephilosophyofthe company is based on reaching and supplying the most comprehensive and optimized solutions for individuals and organizations in various sectors. The company prides itself on integrating the latest and most appropriate approaches and technologies to guarantee sustainable well-adapted and economically efficient solutions for the clients. The company logo is depicted on Figure1.1.



Figure 1.1 – Logo de l'entreprise SMART KOKUSAI

1.3 Organizational Structure of the Host Company

SMART KOKUSAI Ltd, is a company based in Ariana, Tunisia. The main activity of the Company is to provide adaptable and optimized solutions for infrastructure and smart issuesin compliance with the UNDP's SDGs. The company also insures the Audit and the Monitoring of the projects' implementation while providing consulting and training services under the cooperation with the Universities and the research institutes.
Through the use of relevant and advanced analytic tools and data-driven insights, SMART KOKUSAI is exploiting and exploring relevant and viabledatathatmay help decisionmaking and technology / solution appropriation based on considering social, environmental,economic, and legal framework aspects.
The company, adopting a very Japanese philosophy, has three axis related to the engineering and RDI, the consultancy and assistance, and the integration of the ICT and AI tools for sustainable and environmentally sound society. The organization of the company is shown in the Figure 1.2.

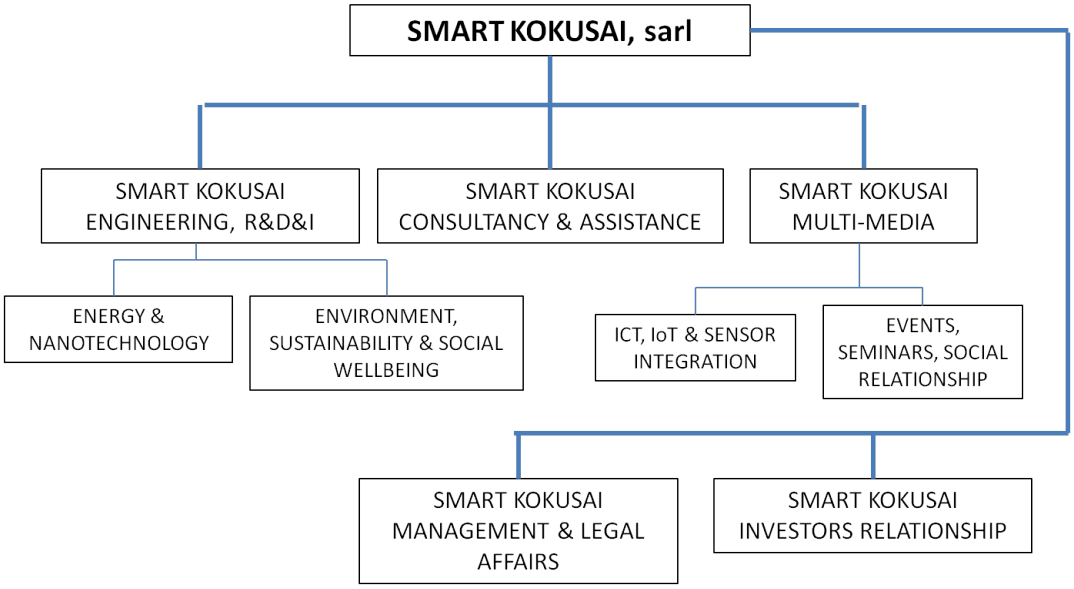


FIGURE 1.2 – OrganizationalStructureof thehost company

1.4 Requirements for the Development of the System

To gain a clearer understanding of the requirements for developing the target system, i first identified the functional, non-functional, and business/strategic requirements.

1.4.1 Functional Requirements

Functional requirements represent the actions that the system performs in response to a request. Hence, it is necessary to clearly identify the system's operational requirements. To do so, I specify the smart tool features as follows :

- **Register CSR activities** – allow companies to record and document their social initiatives.
- **Visualize CSR initiatives** – provide dashboards, statistics, or charts for better understanding.
- **Evaluate CSR impact** – assess and measure the effectiveness of initiatives (reports, indicators).
- **Monitor CSR activities** – track progress and status of ongoing initiatives.
- **Report generation** – systematic analysis and reporting of CSR activities for decision-makers.
- **Support stakeholder engagement** – involve companies, employees, communities, and decision-makers.
- **Ensure alignment with international standards** – integrate indicators consistent with CSR/ESG frameworks.

1.4.2 Nonfunctional requirements

Non-functional requirements are attributes that describe how a system should perform or behave rather than specifying particular functionalities.

- **Accessibility** – mobile-based platform, accessible to organizations of different sizes.
- **User-friendliness** – intuitive interface to encourage adoption by non-technical users.
- **Transparency & accountability** – must provide reliable and clear information for trust-building.
- **Security & privacy** – protect sensitive company and community data (even if not explicitly said, it's implicit when handling CSR reports).

— **Scalability** – able to handle multiple organizations and diverse CSR activities.

— **Sustainability** – designed to support long-term CSR adoption and digital innovation principles.

— **Reliability** – ensure correct and continuous monitoring without frequent downtime.

1.5 Context of the Present Work

The present work focuses on the development of a CSR environmental monitoring tool designed to track and evaluate the company's CSR activities. This tool integrates ICT, IoT, and AI-based technologies to identify, measure, and monitor Key Goal Indicators (KGIs) and Key Performance Indicators (KPIs) related to sustainability objectives, aligned with the Sustainable Development Goals (SDGs) and 6R policies.
My contribution to this project specifically involves the IoT integration and the dashboarding module. The IoT part focuses on connecting sensors and devices to collect real-time environmental data, while the dashboarding component is designed to visualize KGIs and KPIs in an interactive and user-friendly way, enabling managers and employees to efficiently monitor CSR performance.

1.6 Conclusion

The development of the CSR environmental monitoring tool within the SMART KOKUSAI project represents an important step toward integrating sustainable practices into industrial operations. this work contributes to optimizing the monitoring of CSR activities and improving decision-making based on real-time data

Chapitre 2

Internship Context and State of the Art

2.1 Introduction

This chapter presents the theoretical and contextual foundation of the internship, covering CSR and its link to the SDGs, the current IoT-based environmental monitoring technologies, and the objectives and methodology of the conceived smart monitoring solution.

2.2 Corporate Social Responsibility (CSR) and the Sustainable Development Goals (SDGs)

2.2.1 Definition and Significance of CSR

Corporate Social Responsibility (CSR) is a self-regulating business model that ensures a company acts responsibly toward society, stakeholders, and the environment. CSR enables businesses to create positive social and environmental impacts, enhancing long-term profitability, risk management, and brand reputation.

The Environmental Dimension of CSR and the SDGs

The environmental pillar of CSR focuses on reducing a company's ecological footprint. This includes managing waste, reducing greenhouse gas emissions, ensuring sustainable resource consumption, and preventing pollution.
This dimension is directly aligned with several of the UN's Sustainable Development Goals, including :

— **SDG 6 (Clean Water and Sanitation) :** Monitoring water quality and consumption.
— **SDG 7 (Affordable and Clean Energy) :** Optimizing energy efficiency.
— **SDG 11 (Sustainable Cities and Communities) :** Managing waste and pollution in urban environments.
— **SDG 12 (Responsible Consumption and Production) :** Ensuring efficient use of resources and reducing waste.
— **SDG 13 (Climate Action) :** Tracking and reducing carbon emissions.

The Challenge of Measurement and Reporting

A significant challenge for companies is moving from vague commitments to actionable, measurable results. This is where technology plays a pivotal role. Accurate, real-time data is essential for :

— **Monitoring :** Tracking environmental parameters (e.g., energy consumption, waste levels, air/water quality) reliably.
— **Reporting :** Providing transparent, verifiable data for sustainability reports to stakeholders, regulators, and consumers.
— **Improving (Kaizen) :** Using data-driven insights to optimize processes, reduce environmental impact, and achieve continuous improvement.

2.3 State Of The Art

Corporate Social Responsibility (CSR) has become integral to modern business strategies, with organizations striving to align their operations with social, environmental, and ethical standards. However, effectively managing and monitoring CSR initiatives remains a challenge due to the complexity and scale of these activities.

Existing CSR Monitoring Tools

Current CSR software solutions offer a range of features to assist organizations in managing their CSR efforts :

Integrated Platforms :

Comprehensive systems like Bonterra provide centralized management of grantmaking, volunteer programs, employee giving, and impact measurement, eliminating the need for multiple disparate tools.

Mobile Applications : Platforms such as Benevity and Goodera offer mobile-friendly solutions that facilitate employee engagement through features like volunteer tracking, donation matching, and real-time reporting.

Analytics and Reporting : Tools like Metrio and Ecovadis offer advanced analytics capabilities, enabling organizations to assess ESG (Environmental, Social, Governance) performance and generate reports aligned with international standards.

Customization and Scalability : Solutions such as TechCSR and Optimy provide customizable platforms that can be tailored to the specific needs of organizations, ensuring scalability and adaptability.

Limitations of Current Solutions Despite the advancements in CSR software, several limitations persist :

Complexity and Cost :

Many comprehensive CSR platforms are complex to implement and may be cost-prohibitive for small to medium-sized enterprises.

Limited Focus on Social Impact : While many tools offer robust features for environmental and governance tracking, there is often a lack of emphasis on the social aspects of CSR initiatives.

Integration Challenges : Integrating CSR software with existing organizational systems, such as HR and finance tools, can be challenging, leading to data silos and inefficiencies.

The Need for a Smart Mobile Monitoring Tool

Given these limitations, there is a clear need for a smart, user-friendly mobile application that :

Enhances Transparency and Accountability : Provides real-time tracking and reporting of CSR activities, ensuring stakeholders have access to up-to-date information.

Encourages Employee and Community Engagement : Facilitates participation through mobile devices, making it easier for employees and community members to get involved in CSR initiatives.

Aligns with International Standards : Supports compliance with global CSR and ESG frameworks, aiding organizations in meeting regulatory requirements.

Supports Data-Driven Decision Making : Offers analytics capabilities to assess the impact of CSR activities and inform strategic planning.
By addressing these needs, the proposed mobile application aims to streamline CSR monitoring processes, improve stakeholder engagement, and enhance the overall effectiveness of CSR initiatives.

Bridging CSR and ICT

The convergence of the environmental domain (CSR) and Information and Communication Technologies (ICT), specifically IoT, is the core innovation of our project. IoT provides the necessary tools to overcome the traditional challenges of environmental monitoring by enabling :

- **Real-time Data Acquisition :** Moving from manual, periodic checks to continuous, automated data streams.
- **Scalability :** Deploying networks of sensors across multiple locations.
- **Cost-Effectiveness :** Reducing the labor and time required for data collection.
- **Data-Driven Decision Making :** Providing the foundation for meaningful Kaizen and strategic CSR initiatives.

2.4 Objectives of the Conceived Smart Monitoring System

The internship aims to design a smart IoT-based tool to monitor environmental parameters for CSR activities. Key goals include automating data collection, real-time KPI monitoring via a dashboard, simplifying reporting, and demonstrating a functional prototype for future industrial use.

2.5 Work Methodology : Agile-Scrum

We adopted the Agile methodology for the development of our smart monitoring system. Its flexible and iterative nature is perfectly suited to a complex R&D project with evolving requirements.

Why Agile ?

- **Customer Satisfaction :** Delivers functional software increments regularly, ensuring the solution meets the client's needs.
- **Adaptability to Change :** Embraces and manages changing requirements, even late in development.
- **Iterative Development :** The project is broken down into small cycles (sprints), each resulting in a potentially shippable product increment. This allows for continuous testing and feedback.
- **Collaborative Approach :** Promotes constant communication between the development team, stakeholders, and the client.

Scrum Framework

We specifically used the Scrum framework to implement Agile :

- **Roles :** The team is composed of a Scrum Master (facilitator), a Product Owner (representing client/stakeholder interests), and the Development Team.
- **Sprints :** Development occurs in time-boxed iterations (sprints), typically 2-4 weeks long.
- **Artifacts :** We used a **Product Backlog** (a prioritized list of features), a **Sprint Backlog** (tasks for the current sprint), and a **Sprint Review** to demonstrate the increment.
- **Continuous Improvement :** Each sprint ends with a **Sprint Retrospective** to identify what went well and what could be improved in the next iteration.

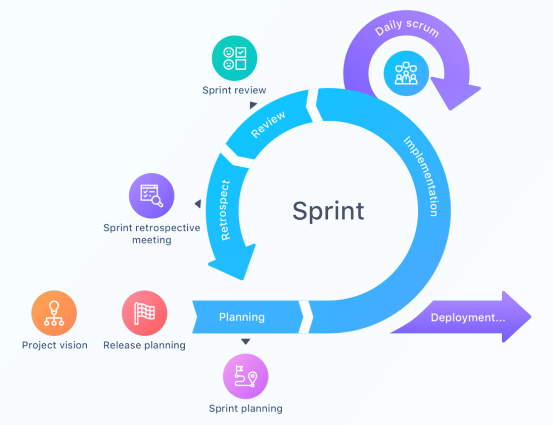


FIGURE 2.1 – The Agile Scrum Management Process

2.6 Conclusion

This chapter highlighted the importance of CSR in achieving the SDGs and the need for reliable environmental data. IoT technology was presented as the solution, with an Agile-Scrum methodology guiding development. The initial prototype monitors temperature and humidity, laying the foundation for a full smart monitoring system. The following chapters detail its design and implementation.

Chapitre 3

Design of the Smart CSR Monitoring System

3.1 Introduction

Following the analysis of requirements and the state of the art presented in the previous chapter, this chapter is dedicated to the design phase of the smart CSR monitoring tool.It breaks down the system's core components : hardware for data acquisition, software for data flow, and the storage and visualization layers. Serving as a blueprint for implementation, this chapter establishes the foundational design for the subsequent development phase.

3.2 System Design and Operation

The Smart CSR Monitoring System is architected as a distributed application following a data-driven approach. Its purpose is to aggregate data from both automated physical sensors and manual user input to compute and visualize Key Performance Indicators (KPIs) for corporate social responsibility. The high-level system design, which facilitates parallel development, is presented in Figure 4.10.

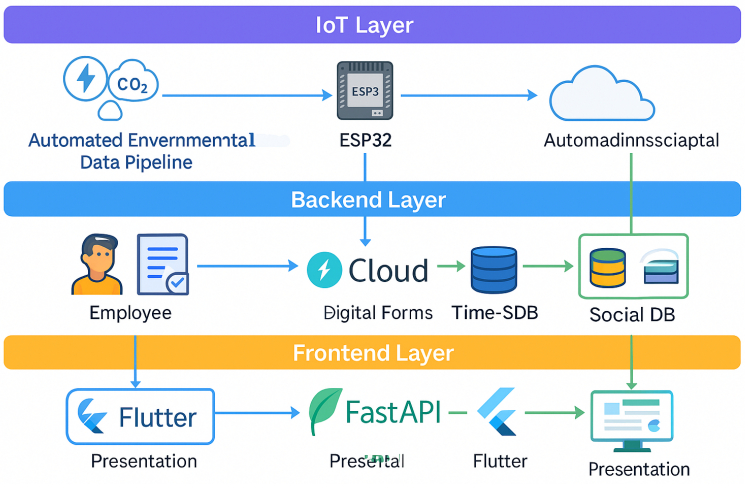


FIGURE 3.1 – High-level architecture of the Smart CSR Monitoring System

The system is structured into three main layers to ensure scalability and clear separation of concerns :

- **IoT Layer :** Responsible for collecting environmental data from physical sensors.
- **Backend Layer :** Manages data storage, processing, and API services.
- **Frontend Layer :** Provides user interfaces for data visualization and interaction.

While the system is designed to integrate both environmental and social responsibility data streams, this work focuses exclusively on the environmental pipeline, which is detailed in the following section.

Automated Environmental Data Pipeline

This pipeline forms the core contribution of this project and is implemented as follows :

1. **Data Acquisition :** Sensors continuously measure environmental KPIs such as energy consumption, water usage, and CO2 levels.
2. **Microcontroller Processing :** An microcontroller acquires and performs initial pre-processing of the sensor data.
3. **Data Aggregation :** A gateway collects data from multiple ESP32 nodes across the network.
4. **Cloud Transmission :** The aggregated data is transmitted to the cloud using a communication protocol.

5. **Data Storage :** Sensor readings are stored in a time-series database optimized for temporal data.
6. **Visualization :** connects to the database to provide professional visualization and dashboard capabilities.

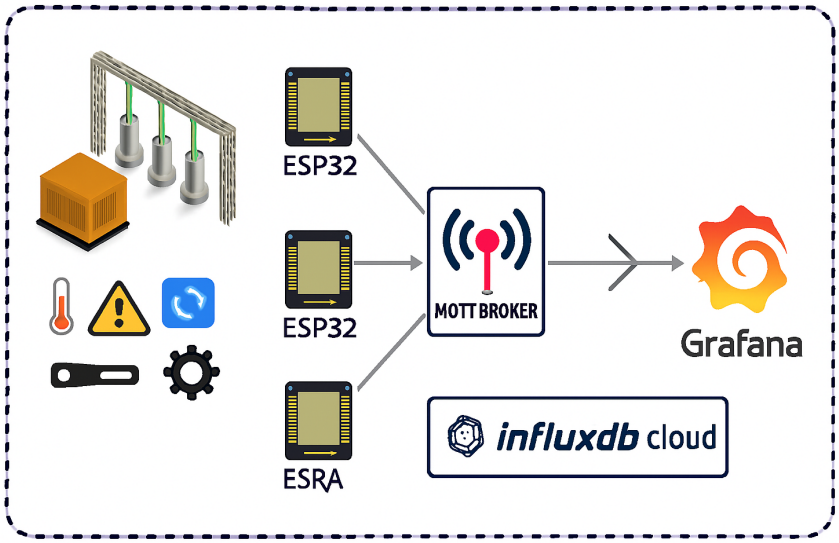


FIGURE 3.2 – Environmental data pipeline : from sensor acquisition to visualization.

Operational Workflow

To facilitate the everyday activities of citizens and the work of CSR/environmental managers, the smart tool was designed with a simple and economical operating principle.
The tool is divided into two main parts : **Hardware** and **Software**, with the goal of providing real-time monitoring of environmental parameters while enabling easy user interaction.

Hardware Part

The hardware part is responsible for collecting real-time environmental data and controlling physical actuators where needed. Our design focuses on measuring key CSR-related metrics such as :

- **Sensors :**
  Energy meters to track electricity consumption.
  Water flow sensors for real-time monitoring of water usage.
  CO2 sensors to monitor emissions and environmental impact.
- **Microcontroller :** responsible for collecting data from all sensors, performing initial processing, and transmitting it securely.

The system also includes a communication module to send collected data to the backend securely. Real-time

Software Part

The software part is responsible for storing, processing, and visualizing the data collected by the sensors. Its main functions include :

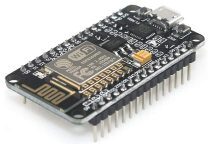
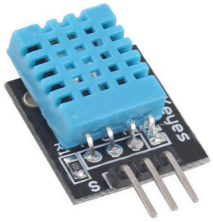
- Store real-time environmental data in a secure database accessible only to authorized personnel.
- Provide dashboards for monitoring energy, water, waste, and CO_2 metrics.
- Generate alerts when specific thresholds are exceeded, helping managers take timely action.

This design ensures that employees and managers can easily monitor environmental performance, make data-driven decisions, and contribute to the company's CSR objectives.
The combination of hardware and software ensures a fully functional prototype that is sustainable, efficient, and aligned with CSR objectives.

selection of components

To implement the basic features of the smart CSR tool, we selected simple yet effective electronic components to achieve the expected results. The table below (Table 3.1) summarizes the materials and equipment used in building the prototype.

TABLE 3.1 – List of Components

Component Name	Figure	Role
ESP32WROOM-32D Wifi + Bluetooth		• **Low Cost** – Suitable for large-scale IoT deployments. • **Built-in Communication** – Supports Wi-Fi and Bluetooth. • **High Functionality** – Converts analog data into digital format.
DHT11 Humidity and Temperature Sensor		• Measures temperature and humidity.
Water Flow Sensor YF-S201		• **Accurate Measurement** – Detects flow rates from 1 to 30 L/min. • **Hall Effect Technology** – Ensures precise pulse generation.
PZEM-004T 100A Energy Meter Module		• Measures voltage, current, power factor, and energy. • Handles up to 100A with split CT. • UART communication for easy integration.
Raspberry Pi 4 Model B (2GB RAM)		• **Compact and Powerful** – Quad-core CPU, 2GB RAM. • **Multi-OS Support** – Works with Raspberry Pi OS, Ubuntu, etc.
Ultrasonic Ranging Module HC-SR04		• Detects waste levels inside bins.

3.3 Technologies Used

For the interface design we used "Figma" which is suitable to build user interfaces (UI) and user experiences (UX). It is a web-based collaborative design tool. It offers tools for experimenting, developing, and exchanging design elements, allowing designers to collaborate in real time. Figma is renowned for its simplicity, adaptability, and availability on several platforms.



FIGURE 3.3 – Logo of Figma

InfluxDB Cloud is a cloud-based time series database service provided by InfluxData. InfluxDB is an open-source time-series database function that provides a full range of features for handling time-series data generated from IoT devices and other sources



FIGURE 3.4 – Logo of InfluxDB

Visualization Tools In this case study, Grafana Cloud is a SaaS observability platform on the cloud managed by Grafana Labs. The platform enables the collection, visualization, and alert notification of observables and logs without the need for a proprietary mechanism. As a fully managed service built on the cloud, it is scalable and does not require users to manage and maintain their own infrastructure.



FIGURE 3.5 – Logo of Grafana

The open-source "Docker" enables the development, distribution, and ad ministration of containerized programs. It makes software packaging and distribution simpler by providing a consistent and isolated environment for programs to run in many computers.



FIGURE 3.6 – Logo of Docker

TheEclipse Mosquitto is an open-source message broker (licensed by EPL/EDL) that implements MQTT protocol versions 5.0, 3.1.1, and 3.1. Mosquitto is lightweight and suitable for use on all devices from low-power single-board computers to complete servers.



FIGURE 3.7 – Logo of Mosquitto

3.4 Flowcharts and Diagrams

The system uses **Role-Based Access Control (RBAC)** and has **three types of users** : **Admin**, **Manager**, and **Employee**.

Global Use Case Diagram

Figure 3.8 illustrates the use case diagram, which describes the key functionalities provided by the system to its different categories of users. This diagram serves as a high-level representation of the interactions between the actors and the system, highlighting how each type of user can access and utilize the features offered. It provides a clear overview of the system's scope, its main services, and the roles of the users involved.

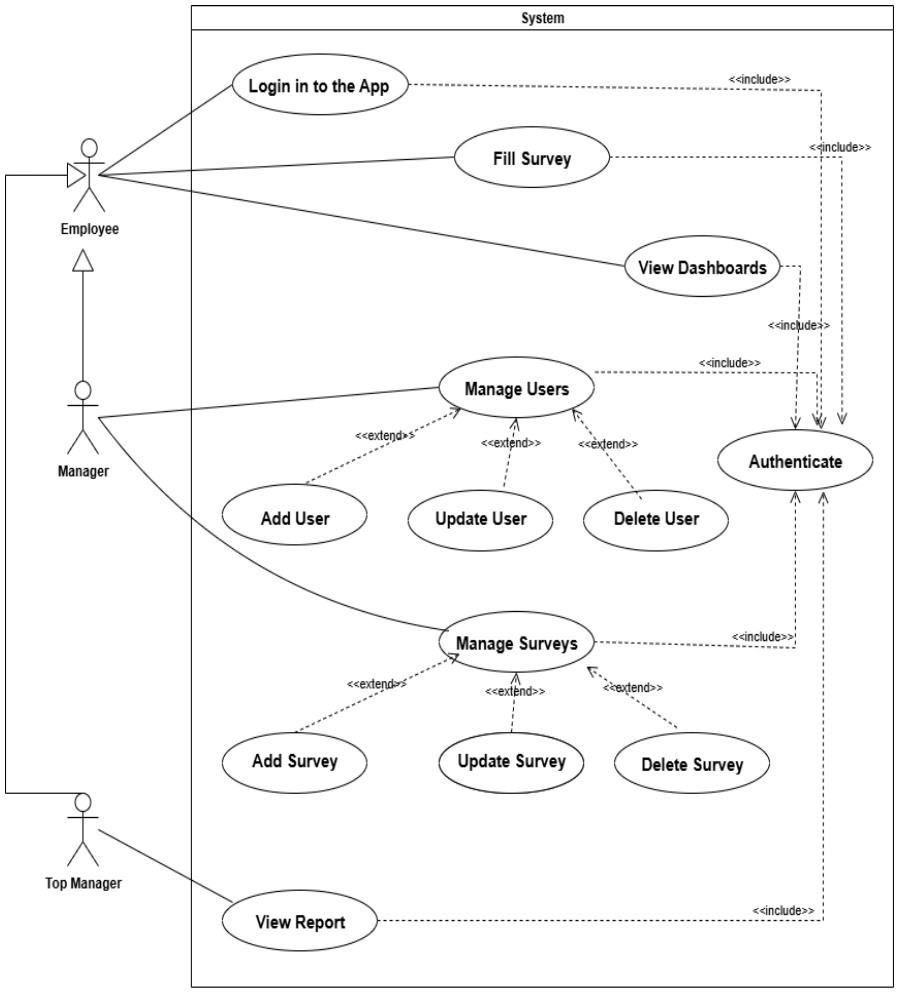


FIGURE 3.8 – Use case Diagram

Actors :

- **Top Manager :** Oversees the overall CSR strategy, approves initiatives, and ensures alignment with the company's vision and international standards.
- **Manager :** Coordinates and supervises CSR activities within their department, monitors progress, and reports results to top management.
- **Employee :** Participates in CSR initiatives, provides feedback on activities, and contributes to the company's social impact through direct engagement.

3.5 Networking

This section presents the networking architecture used in our project, focusing on the communication protocols and the integration of data transmission and storage.

Communication Protocol : MQTT

For real-time communication between devices, we adopted the **MQTT** protocol. It is widely used in IoT systems due to its lightweight and efficient design. The main features of MQTT are :

- **Lightweight and Efficient :** Designed for constrained devices with limited processing power and low bandwidth.
- **Publish-Subscribe Model :** Data publishers send messages to topics, while subscribers receive messages by subscribing to these topics.
- **Asynchronous Communication :** Devices exchange data independently without waiting for immediate responses.
- **Quality of Service (QoS) :** Ensures reliable delivery with three levels.

MQTT was chosen for this project due to its reliability, scalability, and compatibility with IoT environments.

Data Transmission and Storage Integration

Sensor data from **ESP32** nodes is sent via **BLE** to a **Raspberry Pi** gateway, which aggregates and transmits it over **Wi-Fi** using **MQTT over TLS**. The cloud ingests data via an MQTT client and stores it in **InfluxDB** as time-series data.

Role of Each Component

- **BLE** : Low-power sensor-to-gateway communication
- **Wi-Fi** : Gateway-to-cloud data transfer

- **MQTT** : Reliable and efficient messaging
- **InfluxDB** : Time-series data storage

3.6 Working Environment

This section presents the hardware and software environments used during the development and testing of the smart monitoring tool. Two trials were conducted : the first using a personal computer and the second using a Raspberry Pi as the main controller.

Hardware Environment

We used a PC and a Raspberry Pi as the hardware environment for prototyping the HPCS monitoring system. Tables 3.2 and 3.3 describe their specifications.

Table 3.2 – PC Specifications

Component	Details
PC	Desktop 212+991
Processor	Intel Core i5-10265U
RAM	20 GB
Storage	1 TB HDD
Operating System	Windows 10

Table 3.3 – Raspberry Pi Specifications

Component	Details
Raspberry Pi	Raspberry Pi 4 Model B
Processor	Broadcom BCM2837B0
RAM	2 GB
Storage	MicroSD card
Operating System	Raspbian (Linux)

Software Environment

The development was performed using the **Visual Studio Code (VS Code)** IDE, which provided an efficient environment for writing, debugging, and testing code.



FIGURE 3.9 – Visual Studio Code IDE used for development

In addition to the IDE, several tools and libraries were used to facilitate the implementation of the system.
For the ESP32, we used **Espressif's ESP-IDF framework (v3.6.2)** , ensuring full compatibility with the ESP32 hardware.



FIGURE 3.10 – logo ofESP-IDF framework

3.7 Conclusion

This chapter has detailed the architectural design of the smart CSR monitoring system, establishing a clear blueprint for implementation. The proposed structure integrates IoT components for data acquisition, a robust backend for processing and storage, and an intuitive dashboard for visualization. This comprehensive design ensures the system will effectively meet the specified functional and technical requirements.
The following chapter will focus on the practical implementation of this design and the evaluation of the resulting prototype.

Chapitre 4

Implementation and Results

4.1 Introduction

This chapter details the practical implementation of the smart CSR monitoring system. It covers the prototyping of the environmental sensor module, the end-to-end configuration of the data pipeline, and the comprehensive testing and validation procedures undertaken to ensure the system's reliability, accuracy, and effectiveness for CSR reporting.

4.2 Prototyping Process of the Environmental Monitoring Module

Figure 4.1 shows the Administrator Use Case Diagram. Data from various sensors (waste, water, temperature, humidity, and energy) are collected via the ESP32 and transmitted using MQTT to Telegraf, then stored in InfluxDB and visualized in Grafana for real-time CSR environmental monitoring.

Hardware Assembly and Integration

The hardware prototype consisted of environmental sensors, and communication modules for wireless data transmission.

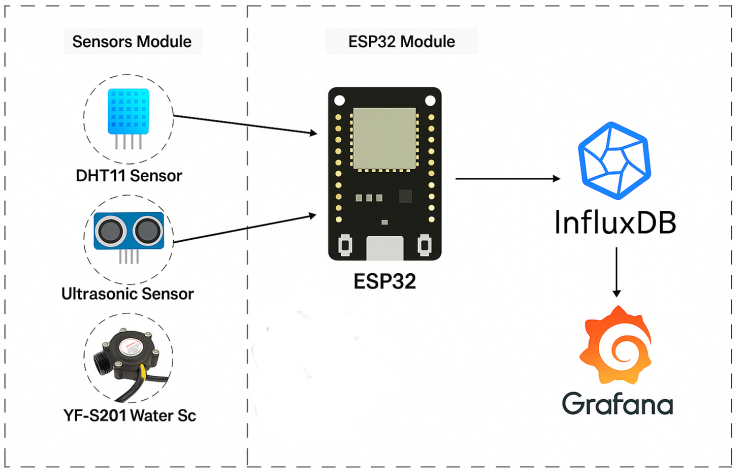


FIGURE 4.1 – Use Case Diagram

4.3 Configuration of the Data Processing Pipeline

During the testing phase, **I** carried out the full configuration of the data collection and visualization pipeline, ensuring that the sensors connected to the ESP32 could reliably transmit environmental data for CSR monitoring.

Telegraf Configuration

I configured **Telegraf** as the data ingestion agent to receive MQTT messages directly from the ESP32 :

- **MQTT Input Plugin :** Defined broker address, topic structure, and authentication parameters to securely subscribe to sensor data topics (waste, water, temperature, humidity, and energy consumption).
- **Data Parsing :** Used the JSON parser to extract individual sensor readings from the payload.
- **Output Plugin :** Configured Telegraf to write parsed data into **InfluxDB** with appropriate measurement names, tags, and timestamps.
- **Buffering :** Enabled output buffering and retry mechanisms to prevent data loss during network interruptions.

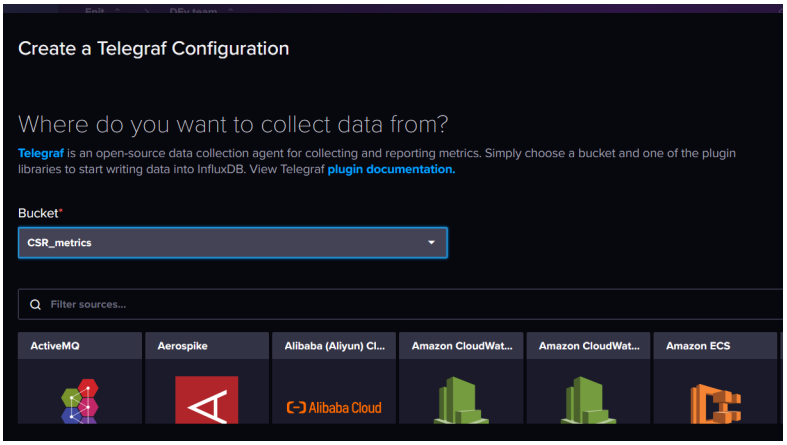


FIGURE 4.2 – Telegraf configuration

InfluxDB Configuration

I set up **InfluxDB** as the time-series database to store all incoming sensor data :

- Created a dedicated database named `csr_metrics` for CSR-related environmental metrics.

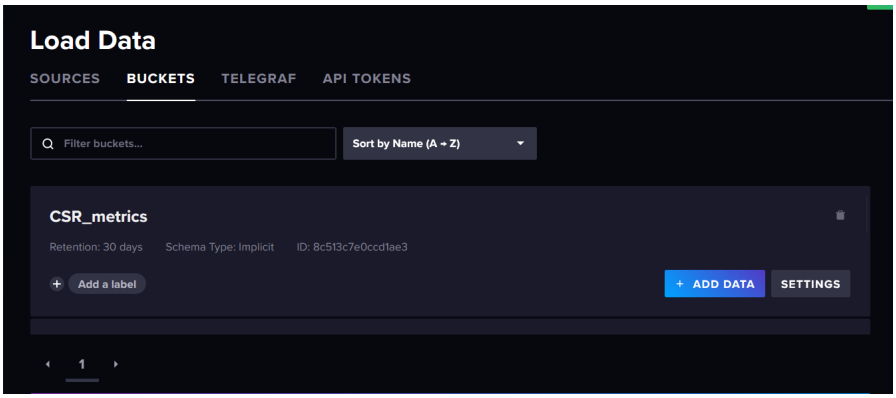


FIGURE 4.3 – Bucket configuration

During testing, I confirmed that the InfluxDB authentication token was properly configured and successfully used by Telegraf to securely write incoming sensor data to the database.

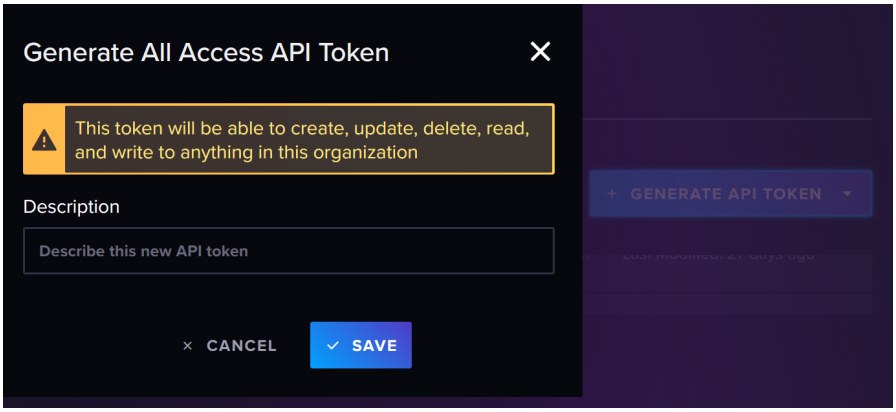


FIGURE 4.4 – Token configuration

Grafana Configuration

I configured **Grafana** to visualize stored data :

- Data Source Connection : I connected Grafana to the csr monitoring database within InfluxDB. This was done by adding a new data source in Grafana's settings and configuring it using the SQL query language to retrieve the sensor data.

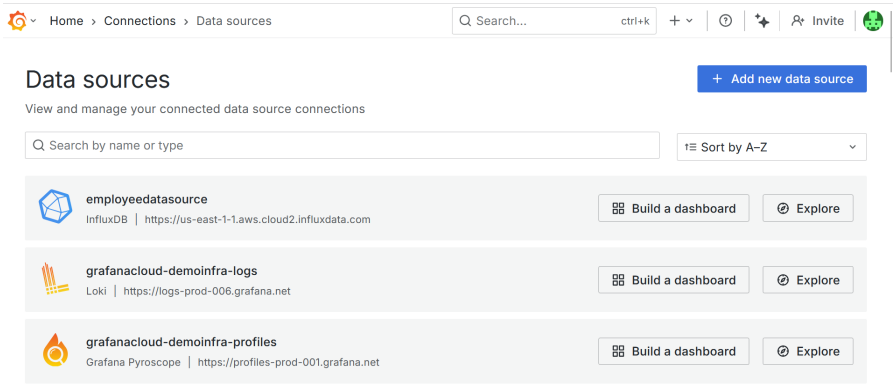


FIGURE 4.5 – Grafana data source configuration panel showing the connection to the InfluxDB database.

- To enable instant status recognition, I set visual thresholds for each parameter. I used a color-coding system where green indicates normal

operation, yellow a warning state, orange a critical state, and red a severe breach of compliance limits. This provides an immediate, at-a-glance understanding of system health.

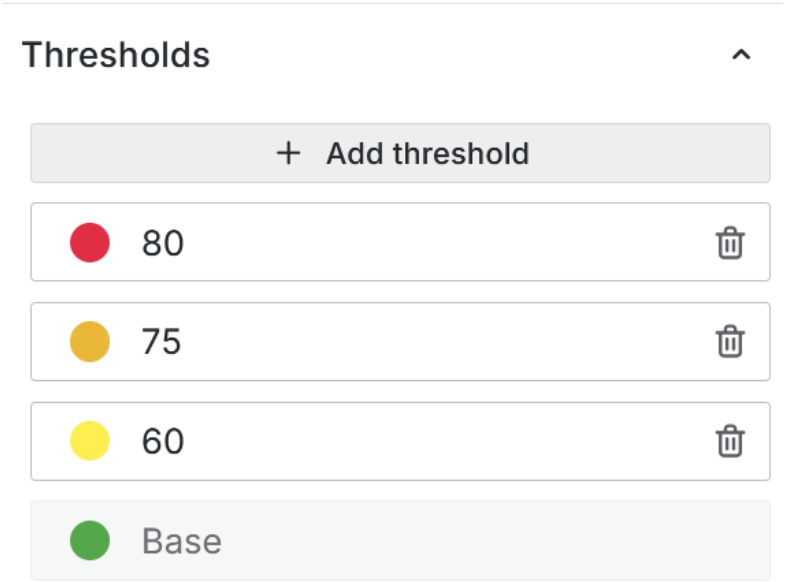


FIGURE 4.6 – thresholds visualization

- configured alert rules based on the same thresholds. For example, if energy consumption exceeds a predefined limit for 5 consecutive minutes, Grafana triggers an alert. This alert is set up to send immediate email notifications to the responsible team, ensuring prompt action to address any CSR compliance issues.

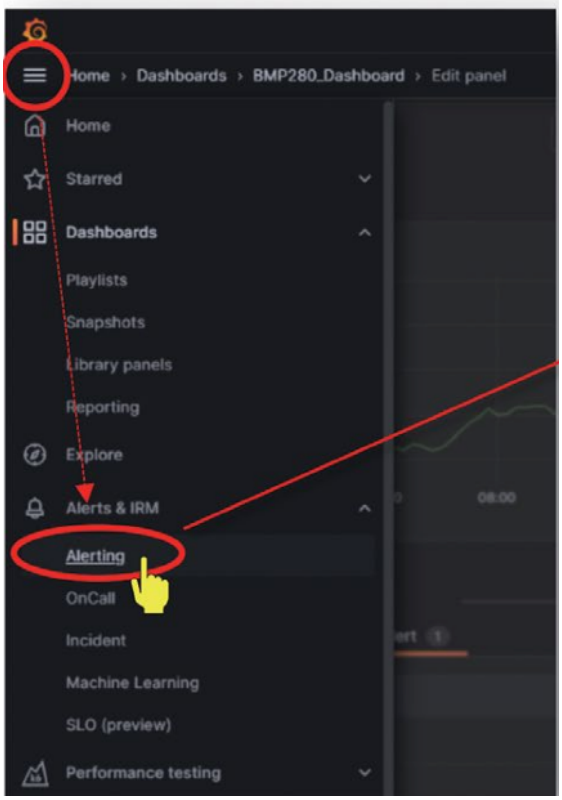


FIGURE 4.7 – Alert configuration

4.4 Technical realization

After completing the configuration of the IoT pipeline, comprehensive end-to-end testing was conducted to verify system stability and data accuracy. The testing phase focused on validating the entire data flow from sensor acquisition to visualization.

4.4.1 Prototype Testing and Validation

Before deploying the sensors in their final locations, we conducted initial testing using a simplified assembly on a breadboard (Figure 4.8). This approach allowed us to verify the functionality of all components and identify any integration issues early in the development process.



FIGURE 4.8 – Testing phase

After connecting the four environmental sensors — the water flow sensor, ultrasonic waste level sensor, DHT11 temperature and humidity sensor, and the energy monitoring module — to the ESP32 microcontroller, the initial programming code was uploaded, and a series of comprehensive functionality tests were conducted. The primary objective of this phase was to verify that each sensor delivered accurate, real-time data and that the ESP32 could simultaneously process and transmit all measurements without interference.
Figure 4.9 illustrates the **waste level detection system**, which utilizes an **ultrasonic sensor** to measure the distance to the waste surface. This distance is then converted into a **fill level percentage**, allowing accurate monitoring of the bin's capacity. The **DHT11 sensor** measures the **ambient temperature and humidity**, providing essential environmental data, while the **water flow sensor** tracks **real-time water consumption** to analyze usage patterns.

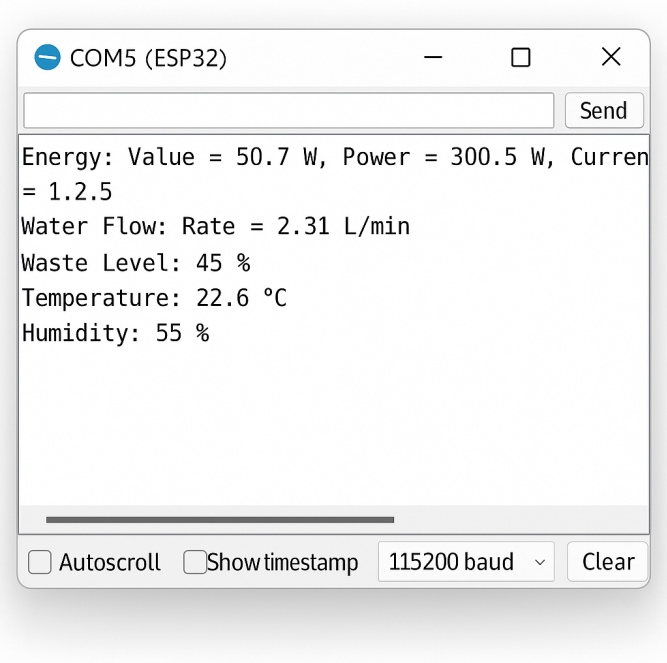


FIGURE 4.9 – Sensor Data Transmitted to the ESP32

4.4.2 Data Transmission Testing

Data Transmission and Pipeline Integrity Test

- **Procedure :** We conducted comprehensive monitoring of the entire data pipeline (ESP32 -> MQTT Broker -> Telegraf -> InfluxDB -> Grafana) over a 24-hour period under normal operational conditions to validate system reliability.
- **Challenge & Solution :** During initial testing, we identified data formatting inconsistencies originating from the ESP32 microcontroller. To address this, we developed and implemented a data validation and parsing layer within Telegraf to standardize the incoming data before storage in InfluxDB.
- **Database Implementation :** We configured InfluxDB to efficiently store and organize the sensor measurements with proper tagging and timestamping. The database structure successfully captures all environmental parameters including water flow, waste levels, temperature, and humidity with their respective measurement units.

With all sensors functioning correctly and providing accurate environmental measurements, we established a robust data transmission system from the ESP32

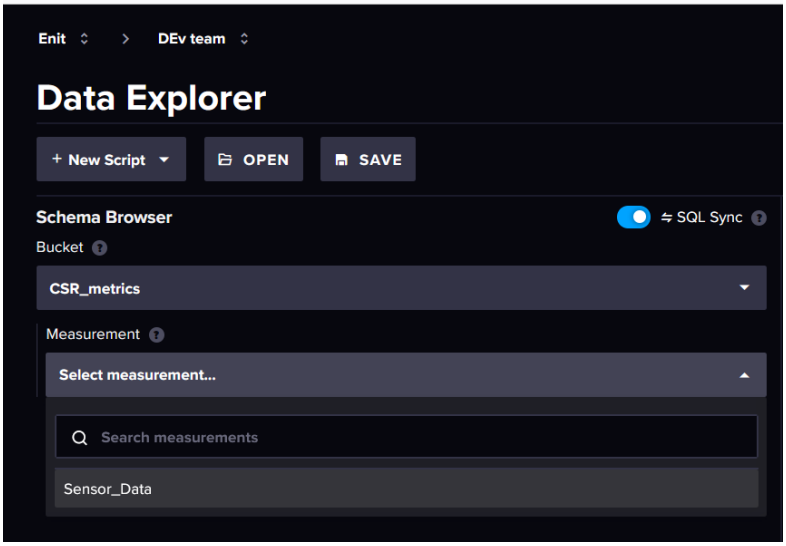


FIGURE 4.10 – InfluxDB database interface showing real-time storage of sensor measurements with proper data structuring and timestamp organization

microcontroller to our central monitoring platform. The implemented pipeline successfully handles multiple critical functionalities :

- **Real-time Data Collection :** We developed continuous acquisition of measurements from all environmental sensors with minimal latency
- **Structured Database Storage :** We implemented efficient data organization in InfluxDB with appropriate tagging for optimized querying and analytical processing

Testing Outcome : The complete data pipeline demonstrates stable operation with consistent real-time data flow from sensor level to visualization interfaces. All measurements are properly stored in InfluxDB and accessible for dashboard display and analytical purposes.

4.4.3 Data Visualization and RBAC Dashboard System

The smart monitoring tool implements a Role-Based Access Control (RBAC) system with three distinct user tiers, as established in the design flowchart. Each role accesses tailored dashboards with appropriate data granularity and control capabilities.

- **Employee Dashboard :** Provides basic operational data and personal consumption metrics for individual awareness and daily monitoring.

- **Manager Dashboard :** Offers department-level analytics with KPI-specific dashboards (e.g., Water Dashboard, Energy Dashboard) featuring consumption thresholds and performance indicators.
- **Top Manager Dashboard :** Delivers global organizational overview with cross-departmental analytics and strategic insights.

Dashboard Implementation and Testing

In addition, we have developed enhanced graphical visualization options for data representation as an alternative to basic tabular displays. The system incorporates multiple dashboard types, each designed for specific user roles and monitoring requirements.
Furthermore, we have implemented specialized sensor data processing to account for measurement variations. The filling level sensor, while maintaining an acceptable error margin as illustrated in the calibration metrics, provides efficient and sufficiently accurate data for operational decision-making. This balance between precision and practicality ensures reliable monitoring without compromising system responsiveness.

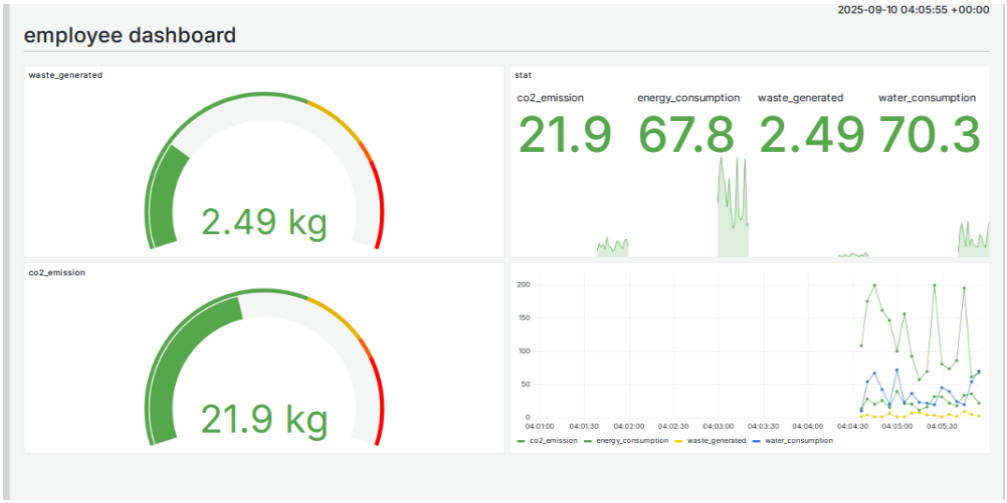


FIGURE 4.11 – Employee-level dashboard showing individual consumption metrics and basic operational data

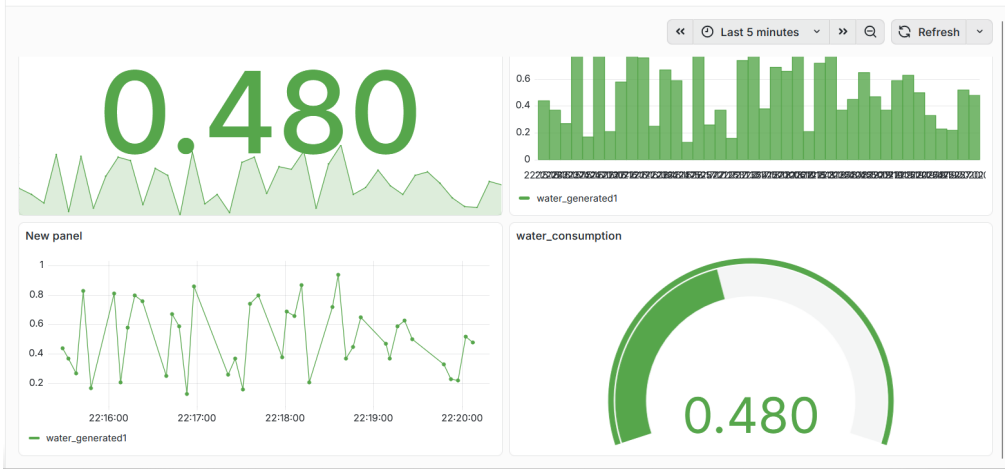


FIGURE 4.12 – Manager-level Water Dashboard displaying consumption patterns, thresholds, and departmental water usage analytics

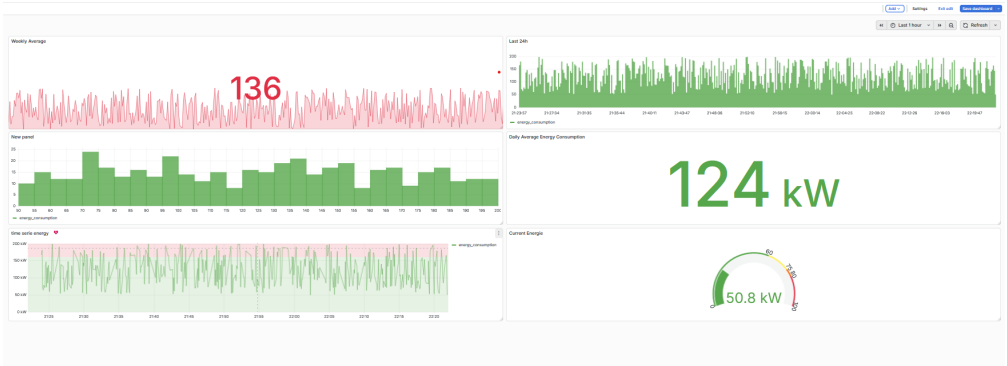


FIGURE 4.13 – Manager-level Energy Dashboard showing energy consumption metrics, usage trends, and efficiency indicators

Testing Outcome : The RBAC dashboard system operates reliably with real-time data updates across all user levels. The graphical visualizations successfully enhance data comprehension, while the sensor accuracy remains within acceptable operational parameters for effective monitoring and decision-making.

4.5 Additional Tasks and Suggestions

Automating Water and Energy Invoice Data Extraction Using OCR

In the context of developing a **smart tool for monitoring CSR activities**, one of the major challenges faced by the company was the **manual processing** of

water and electricity invoices. Traditionally, employees had to **read each invoice, extract the data manually**, and enter it into spreadsheets or databases. This process was **time-consuming, error-prone, and inefficient**.
To overcome this limitation, we integrated an **OCR (Optical Character Recognition)**-based solution to **automatically extract relevant data** from PDF invoices, such as :

- **Energy consumption** (kWh)
- **Water consumption** (m^3)
- **Total cost** (DTN)
- **CO2 emissions estimation**

Architecture of the Solution

1. **Upload Invoice** → The user uploads a PDF invoice through the FastAPI-based web interface.
2. **OCR Processing** → The system converts each PDF page into images, preprocesses them, and uses **Tesseract OCR** to extract text.
3. **Data Parsing** → Regular expressions are applied to extract **consumption**, **cost**, and other key metrics.
4. **Database Storage** → The structured data is stored in **InfluxDB**.
5. **Visualization** → The extracted metrics are displayed on **Grafana dashboards** for real-time monitoring.
   This approach enables the **automation of CSR data collection**, making reporting **faster, more accurate, and more efficient**.

Enhancements in Backend Security and Role-Based Access Control (RBAC)

To ensure **secure access** to the smart monitoring tool, we implemented a **Role-Based Access Control (RBAC)** system in the backend. The system manages users according to three roles :

- **Employee** – Can view and upload invoices related to CSR activities.
- **Manager** – Can review employee submissions and generate reports.
- **Top Manager** – Full administrative privileges, including user management and system configuration.

The RBAC implementation includes :

- **Secure login** with username and password authentication.
- **Role-based restrictions** on actions and data visibility.
- **Integration with FastAPI** for backend access control.

Enterprise Visits and Reference Data Collection

To ensure the tool meets real-world needs, we conducted targeted **on-site visits** to enterprises intended to use the monitoring tool. During these visits, we :

- Held **brainstorming sessions** with enterprise stakeholders to better understand their CSR reporting requirements and operational constraints.
- Collected practical insights to guide the **design of the tool**, including data fields, reporting formats, and user workflows.

Additionally, we visited **SONEDE** and **STEG** to gather reference information such as :

- Typical thresholds for **water and energy consumption**.
- Guidelines and strategies related to the **Sustainable Development Goals (SDGs)** in Tunisia.

This combined approach ensured that both the **OCR data extraction** and **RBAC features** are aligned with actual enterprise practices and national benchmarks.

4.6 Conclusion

The prototyping and testing phase successfully demonstrated the functionality and reliability of the Environmental Monitoring Module. All system components, from hardware sensors to cloud visualization, performed within expected parameters and met the design requirements. The system achieved excellent performance in terms of data accuracy, transmission reliability, and visualization capabilities. The successful implementation of this module provides a solid foundation for integration with the broader Smart CSR system and demonstrates the feasibility of using IoT technology for environmental monitoring in corporate social responsibility applications. The lessons learned and solutions developed during this phase will inform future iterations and deployments of the system.

General Conclusion

This report presented the design and implementation of a **smart environmental monitoring tool** developed within the framework of Corporate Social Responsibility (CSR). Conducted during my internship at **SMART KOKUSAI**, the project focused on addressing the lack of effective digital systems for tracking and evaluating environmental performance within organizations.
The developed solution integrates **IoT sensors** to monitor key environmental indicators , providing real-time data collection and visualization through an intuitive dashboard. Additionally, the system leverages **AI-based Optical Character Recognition (OCR)** to automate the extraction of data from invoices, thus enhancing accuracy and efficiency in CSR reporting.
This tool contributes to promoting **sustainability, transparency, and data-driven decision-making** by enabling companies to assess and optimize their environmental impact. The positive results obtained demonstrate the system's potential to support organizations in aligning with **Sustainable Development Goals (SDGs)** and improving their ecological footprint.

Bibliographie

[1] L. Li and J. Y. Li, "Corporate social responsibility and society 5.0 : Assessing consumer awareness, loyalty, and trust toward socially responsible organizations," *Sustainability*, vol. 16, no. 17, p. 7862, 2024.

[2] S. Jose and K. Venkitachalam, "A matrix model towards csr–moving from one size fit approach," *Journal of Strategy and Management*, vol. 12, no. 2, pp. 243–255, 2019.

[3] K. Tschimmel, "Design thinking as an effective toolkit for innovation," in *ISPIM Conference Proceedings*. The International Society for Professional Innovation Management (ISPIM), 2012, p. 1.

[4] H. Xiao, A. Al Mamun, M. N. H. Reza, X. Lin, and Q. Yang, "Modeling the significance of corporate social responsibility on green capabilities and sustainability performance," *Heliyon*, vol. 10, no. 19, 2024.

[5] W. Zhou, J. Zhu, and C. Zhang, "Impact of corporate social responsibility on carbon emission reduction in supply chains," *Chinese Management Studies*, vol. 18, no. 2, pp. 454–478, 2024.

[6] J. Weinzettel and R. Wood, "Environmental footprints of agriculture embodied in international trade : sensitivity of harvested area footprint of chinese exports," *Ecological Economics*, vol. 145, pp. 323–330, 2018.

[7] N. R. N. A. Rashid, S. A. Khalid, and N. I. A. Rahman, "Environmental corporate social responsibility (ecsr) : Exploring its influence on customer loyalty," *Procedia Economics and Finance*, vol. 31, pp. 705–713, 2015.

[8] A. Janes, A. Sillitti, and G. Succi, "Effective dashboard design," *Cutter IT Journal*, vol. 26, no. 1, pp. 17–24, 2013.

[9] R. B. Belgacem, A. B. Amar, and V. Brescia, "Analyzing the relationship between banking performance and csr in the tunisian context : a comparative study of conventional and islamic banks," *European Journal of Islamic Finance*, vol. 11, no. 2, pp. 1–15, 2024.